sponses with low TRACE score: only exhibits hacking behaviour under the full CoT. We leave empirical investigation on this research question for future work.

Overthinking Overthinking is an issue that could inflate the TRACE score, because the model may produce overly long reasoning traces on easy problems even when it already has the correct internal belief during CoT (Zhang et al., 2025). If the model learns this overthinking behavior during RL training, it may tend to overthink on samples in general, which raises the TRACE score as a whole. Therefore, one possible solution is to calibrate the TRACE score by comparing the TRACE scores of RL-trained model with the initial model on a set of clean questions without loopholes, to measure the extent of score inflation caused by overthinking. This could help calibrate the hacking detection threshold. We leave this calibration against overthinking to future work.

Reasoning Effort In this paper, we use the Truncated Reasoning AUC score as a proxy for the model's relative internal reasoning effort, compared to what is presented in the CoT. We also encourage future work to design other approaches to measure the model's internal reasoning effort, which may be more computationally efficient. One recent work by Chen et al. (2026) uses the ratio of "deep think tokens" as a measure of reasoning effort, where deep think tokens are defined as tokens whose predictions undergo sustained revision in the deep layers before converging. We are interested in seeing whether alternative measures of internal reasoning effort can also effectively detect implicit reward hacking.

Limitation Our experiments focus primarily on reward-model loopholes learned during training; we did not exhaustively study cases in which a non-hacking policy at inference time can already recognize and exploit an in-context hint. TRACE uses the initial-policy TRACE score as a detection threshold; if the initial policy already exhibits hacking behavior on some samples by hacking in-context, this raises the baseline and can reduce sensitivity. Practical mitigations include (i) calibrating the threshold on a small curated validation set of examples believed to be free of loopholes, or (ii) adopting robust statistical thresholds (e.g., percentile-based cutoffs or mixed-checkpoints baselines). We leave the systematic evaluation of these mitigation strategies to future work.

Conclusion

We present TRACE as a promising method for detecting reward hacking in reasoning tasks, when the CoT is not verbalizing the hacking intention and is hard to catch by a text-based monitor. We demonstrate the effectiveness of the method in math and code tasks across two different types of loopholes: in-context loopholes and reward model loopholes. We also showcase how we can discover unknown loopholes from a dataset by clustering with TRACE scores. We hope our method provides a new perspective for reward hacking detection and inspires follow-up research in AI control and oversight.

Acknowledgments

We thank members of the ML2 group at NYU for their inputs at various stages of the project. XW and BP are supported by ERC Consolidator Grant DIALECT 101043235. NJ, RA and HH are supported by the AI Safety Fund and Open Philanthropy. This work is supported in part through the NYU IT High Performance Computing resources, services, and staff expertise. We also thank the LMU-NYU Research Cooperation Program for supporting XW's research stay at NYU.

References

Arash Ahmadian, Chris Cremer, Matthias Gallé, Marzieh Fadaee, Julia Kreutzer, Olivier Pietquin, Ahmet Üstün, and Sara Hooker. Back to basics: Revisiting reinforce style optimization for learning from human feedback in llms, 2024. URL `https://arxiv.org/abs/2402.14740`.

AI@Meta. Llama 3 model card. 2024. URL `https://github.com/meta-llama/llama3/blob/main/MODEL_CARD.md`.

Alon Albalak, Duy Phung, Nathan Lile, Rafael Rafailov, Kanishk Gandhi, Louis Castricato, Anikait Singh, Chase Blagden, Violet Xiang, Dakota Mahan, and Nick Haber. Big-math: A large-scale, high-quality math dataset for reinforcement learning in language models, 2025. URL https://arxiv.org/abs/2502.17387.

Iván Arcuschin, Jett Janiak, Robert Krzyzanowski, Senthooran Rajamanoharan, Neel Nanda, and Arthur Conmy. Chain-of-thought reasoning in the wild is not always faithful. *arXiv preprint arXiv:2503.08679*, 2025.

Bowen Baker, Joost Huizinga, Leo Gao, Zehao Dou, Melody Y. Guan, Aleksander Madry, Wojciech Zaremba, Jakub Pachocki, and David Farhi. Monitoring reasoning models for misbehavior and the risks of promoting obfuscation, 2025. URL https://arxiv.org/abs/2503.11926.

Sam Bowman, Jeeyoon Hyun, Ethan Perez, Edwin Chen, Craig Pettit, Scott Heiner, Kamilė Lukosiūtė, Amanda Askell, Andy Jones, Anna Chen, Anna Goldie, Azalia Mirhoseini, Cameron McKinnon, Chris Olah, Daniela Amodei, Dario Amodei, Dawn Drain, Dustin Li, Eli Tran-Johnson, John Kernion, Jamie Kerr, Jared Mueller, Jeffrey Ladish, Joshua D. Landau, Kamal Ndousse, Liane Lovitt, Nelson Elhage, Nicholas Schiefer, Nicholas Joseph, Noem'i Mercado, Nova Dassarma, Robin Larson, Sam McCandlish, Sandip Kundu, Scott Johnston, Shauna Kravec, Sheer El Showk, Stanislav Fort, Timothy Telleen-Lawton, Tom B. Brown, T. J. Henighan, Tristan Hume, Yuntao Bai, Zac Hatfield-Dodds, Benjamin Mann, and Jared Kaplan. Measuring progress on scalable oversight for large language models. *ArXiv*, abs/2211.03540, 2022. URL https://api.semanticscholar.org/CorpusID:253384413.

Wei-Lin Chen, Liqian Peng, Tian Tan, Chao Zhao, Blake JianHang Chen, Ziqian Lin, Alec Go, and Yu Meng. Think deep, not just long: Measuring llm reasoning effort via deep-thinking tokens, 2026. URL https://arxiv.org/abs/2602.13517.

Yanda Chen, Joe Benton, Ansh Radhakrishnan, Jonathan Uesato, Carson Denison, John Schulman, Arushi Somani, Peter Hase, Misha Wagner, Fabien Roger, et al. Reasoning models don't always say what they think. *arXiv preprint arXiv:2505.05410*, 2025.

Carson E. Denison, Monte Stuart MacDiarmid, Fazl Barez, David Kristjanson Duvenaud, Shauna Kravec, Samuel Marks, Nicholas Schiefer, Ryan Soklaski, Alex Tamkin, Jared Kaplan, Buck Shlegeris, Samuel R. Bowman, Ethan Perez, and Evan Hubinger. Sycophancy to subterfuge: Investigating reward-tampering in large language models. *ArXiv*, abs/2406.10162, 2024. URL https://api.semanticscholar.org/CorpusID:270521305.

Scott Emmons, Erik Jenner, David K. Elson, Rif A. Saurous, Senthooran Rajamanoharan, Heng Chen, Irhum Shafkat, and Rohin Shah. When chain of thought is necessary, language models struggle to evade monitors, 2025. URL https://arxiv.org/abs/2507.05246.

Jiazhan Feng, Shijue Huang, Xingwei Qu, Ge Zhang, Yujia Qin, Baoquan Zhong, Chengquan Jiang, Jinxin Chi, and Wanjun Zhong. Retool: Reinforcement learning for strategic tool use in llms. *ArXiv*, abs/2504.11536, 2025. URL https://api.semanticscholar.org/CorpusID:277824366.

Shibo Hao, Sainbayar Sukhbaatar, DiJia Su, Xian Li, Zhiting Hu, Jason Weston, and Yuandong Tian. Training large language models to reason in a continuous latent space. *arXiv preprint arXiv:2412.06769*, 2024.

Dan Hendrycks, Steven Basart, Saurav Kadavath, Mantas Mazeika, Akul Arora, Ethan Guo, Collin Burns, Samir Puranik, Horace He, Dawn Xiaodong Song, and Jacob Steinhardt. Measuring coding challenge competence with apps. *ArXiv*, abs/2105.09938, 2021. URL https://api.semanticscholar.org/CorpusID:234790100.

Jacob Kahn. Repo state loopholes during agentic evaluation. https://github.com/SWE-bench/SWE-bench/issues/465, September 2025.

Cassidy Laidlaw, Shivam Singhal, and Anca Dragan. Correlated proxies: A new definition and improved mitigation for reward hacking. In *International Conference on Learning Representations*, 2024. URL https://api.semanticscholar.org/CorpusID:268248462.

Tamera Lanham, Anna Chen, Ansh Radhakrishnan, Benoit Steiner, Carson Denison, Danny Hernandez, Dustin Li, Esin Durmus, Evan Hubinger, Jackson Kernion, et al. Measuring faithfulness in chain-of-thought reasoning. *arXiv preprint arXiv:2307.13702*, 2023.

Jack Lindsey, Wes Gurnee, Emmanuel Ameisen, Brian Chen, Adam Pearce, Nicholas L. Turner, Craig Citro, David Abrahams, Shan Carter, Basil Hosmer, Jonathan Marcus, Michael Sklar, Adly Templeton, Trenton Bricken, Callum McDougall, Hoagy Cunningham, Thomas Henighan, Adam Jermyn, Andy Jones, Andrew Persic, Zhenyi Qi, T. Ben Thompson, Sam Zimmerman, Kelley Rivoire, Thomas Conerly, Chris Olah, and Joshua Batson. On the biology of a large language model. *Transformer Circuits Thread*, 2025. URL `https://transformer-circuits.pub/2025/attribution-graphs/biology.html`.

Yihong Liu, Raoyuan Zhao, Hinrich Schütze, and Michael A Hedderich. Large reasoning models are (not yet) multilingual latent reasoners. *arXiv preprint arXiv:2601.02996*, 2026.

METR. Cot may be highly informative despite "unfaithfulness". `https://metr.org/blog/2025-08-08-cot-may-be-highly-informative-despite-unfaithfulness/`, August 2025a.

METR. Recent frontier models are reward hacking. `https://metr.org/blog/2025-06-05-recent-reward-hacking/`, 06 2025b.

OpenAI. Sycophancy in gpt-4o: what happened and what we're doing about it. `https://openai.com/index/sycophancy-in-gpt-4o/`, April 2025.

Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke E. Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Francis Christiano, Jan Leike, and Ryan J. Lowe. Training language models to follow instructions with human feedback. *ArXiv*, abs/2203.02155, 2022. URL `https://api.semanticscholar.org/CorpusID:246426909`.

Jacob Pfau, William Merrill, and Samuel R. Bowman. Let's think dot by dot: Hidden computation in transformer language models. In *First Conference on Language Modeling*, 2024. URL `https://openreview.net/forum?id=NikbrdtYvG`.

Fabien Roger and Ryan Greenblatt. Preventing language models from hiding their reasoning. *arXiv preprint arXiv:2310.18512*, 2023.

Sakana AI. The ai cuda engineer: Agentic cuda kernel discovery, optimization and composition – limitations and bloopers. `https://web.archive.org/web/20250221120954/https://sakana.ai/ai-cuda-engineer/#:%CB%9C:text=%20Limitations%20and%20Bloopers`, February 2025.

Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, YK Li, Yang Wu, et al. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.

Joar Skalse, Nikolaus H. R. Howe, Dmitrii Krasheninnikov, and David Krueger. Defining and characterizing reward hacking. *ArXiv*, abs/2209.13085, 2022. URL `https://api.semanticscholar.org/CorpusID:252545256`.

Miles Turpin, Julian Michael, Ethan Perez, and Samuel R. Bowman. Language models don't always say what they think: Unfaithful explanations in chain-of-thought prompting. In *Thirty-seventh Conference on Neural Information Processing Systems*, 2023. URL `https://openreview.net/forum?id=bzs4uPLXvi`.

Miles Turpin, Andy Arditi, Marvin Li, Joe Benton, and Julian Michael. Teaching models to verbalize reward hacking in chain-of-thought reasoning. *arXiv preprint arXiv:2506.22777*, 2025.

Chulin Xie, Yangsibo Huang, Chiyuan Zhang, Da Yu, Xinyun Chen, Bill Yuchen Lin, Bo Li, Badih Ghazi, and Ravi Kumar. On memorization of large language models in logical reasoning. *arXiv preprint arXiv:2410.23123*, 2024.